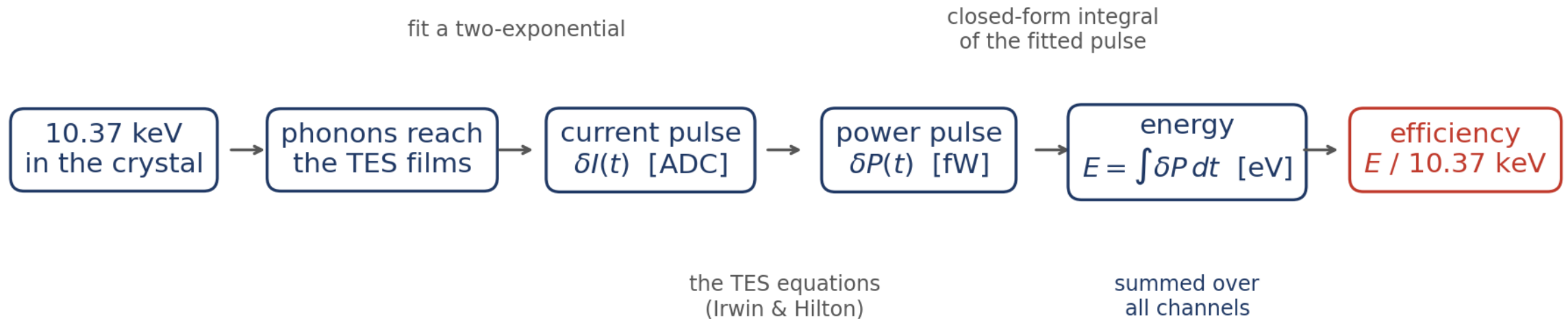


Phonon collection efficiency of Z7

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github.com/ZhihengLi0/urop → [snolab/collection_efficiency/](https://github.com/ZhihengLi0/urop/tree/main/snolab/collection_efficiency/)

The Main Idea



For detectors operated at 0 V (no NTL effect):

Collection efficiency = energy absorbed by the TESs, summed over all channels $\div$ the 10.37 keV of the event

Sample: Z7, the K-line events of the Ge activation run, from SNOLAB R4 (2207 events, 30 series, all 11 channels).

The formula, and the two numbers it needs

Method 1 of the CDMS collection-efficiency note

$$\delta P(t) = I_0 (R_L - R_0) \delta I(t) + 2 R_L (\delta I(t))^2$$

linear term

quadratic term

$$E = \int \delta P(t) dt = I_0 (R_L - R_0) A (\tau_f - \tau_r) + 2 R_L A^2 \left(\frac{\tau_f + \tau_r}{2} - \frac{2 \tau_f \tau_r}{\tau_f + \tau_r} \right)$$

for the fitted pulse $\delta I(t) = A [e^{-t/\tau_f} - e^{-t/\tau_r}]$ integrated from $-\infty$ to $+\infty$: zero before it starts, decays to zero after

$$\text{Z7 PBS1: } I_0 = -18.07 \mu\text{A}, \quad R_0 = 44.16 \text{ m}\Omega, \quad R_L = R_p + R_{sh} = 14.24 \text{ m}\Omega + 5.00 \text{ m}\Omega = 19.24 \text{ m}\Omega$$

$$\text{linear coefficient: } I_0 (R_L - R_0) = (-18.07 \mu\text{A}) \times (19.24 \text{ m}\Omega - 44.16 \text{ m}\Omega) = 0.45 \mu\text{V}$$

$$\text{quadratic coefficient: } 2 R_L = 2 \times 19.24 \text{ m}\Omega = 38.5 \text{ m}\Omega = 0.0385 \Omega$$

δP = the change of Joule heating when the current moves by δI : a linear term and a small quadratic term.

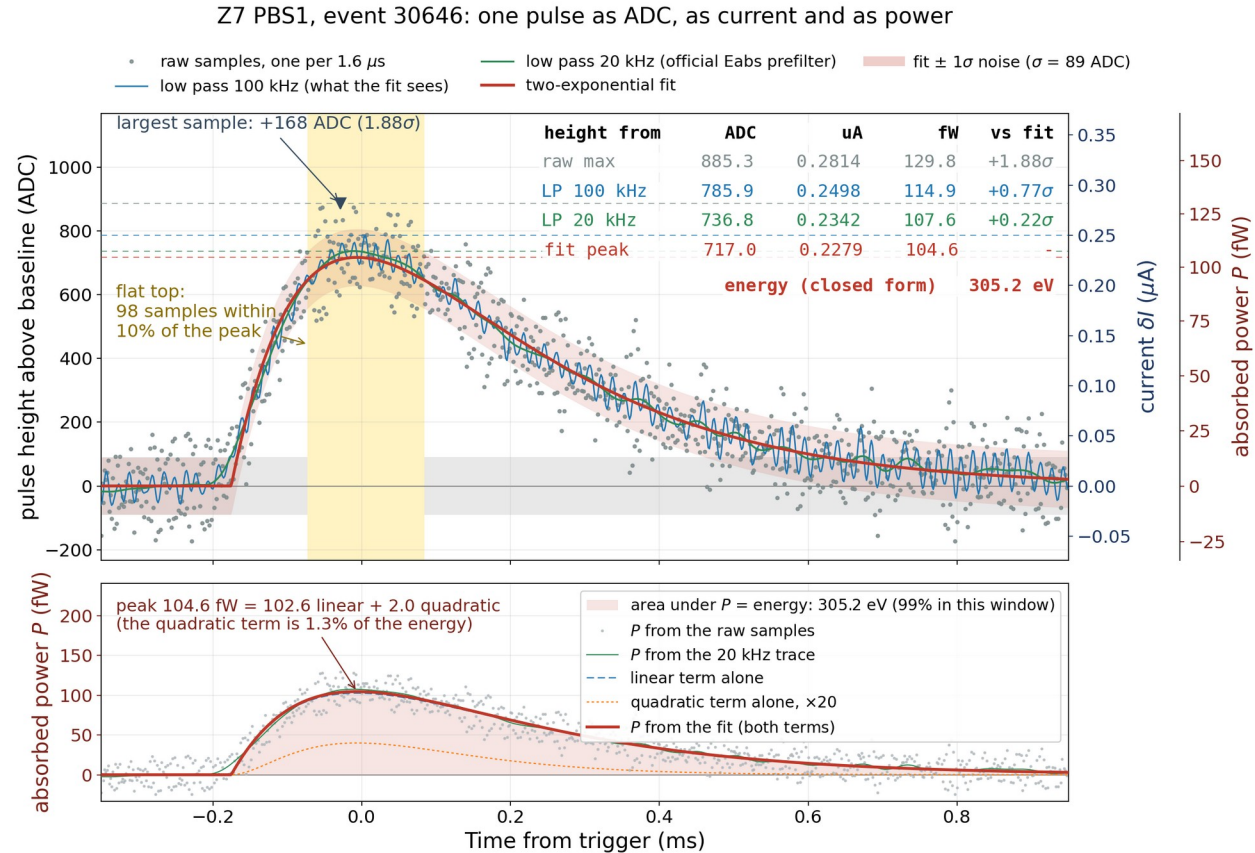
I_0 : TES current at the bias point · R_0 : TES resistance at the bias point · $R_L = R_p + R_{sh}$: load resistance (R_p parasitic, R_{sh} shunt) · δI : change in current, from the trace

The note also gives Method 2: $\delta P = I_0 (R_L - R_0) \cdot \delta I - R_L \cdot (\delta I)^2$ in our sign convention — same linear term, only the quadratic term differs, 1.7 % less energy (PBS1, median of 15 events).

Source: Antoine Rehberg, CDMS wiki, Collection Efficiency Analysis — confluence.slac.stanford.edu/spaces/CDMS/pages/542577254

Example: one pulse

Z7 PBS1, event 30646: the same trace as ADC counts, as current, as power



baseline = mean of bins 93-15758 = 28667.7 ADC, subtracted | 1 ADC = 0.3179 nA

$\delta P = I_0(R_L - R_0)\delta I + c_2 R_L (\delta I)^2$ (method1, $c_2 = 2$), with $I_0 = -18.069 \mu$ A, $R_0 = 44.159$ m Ω , $R_L = 19.240$ m Ω

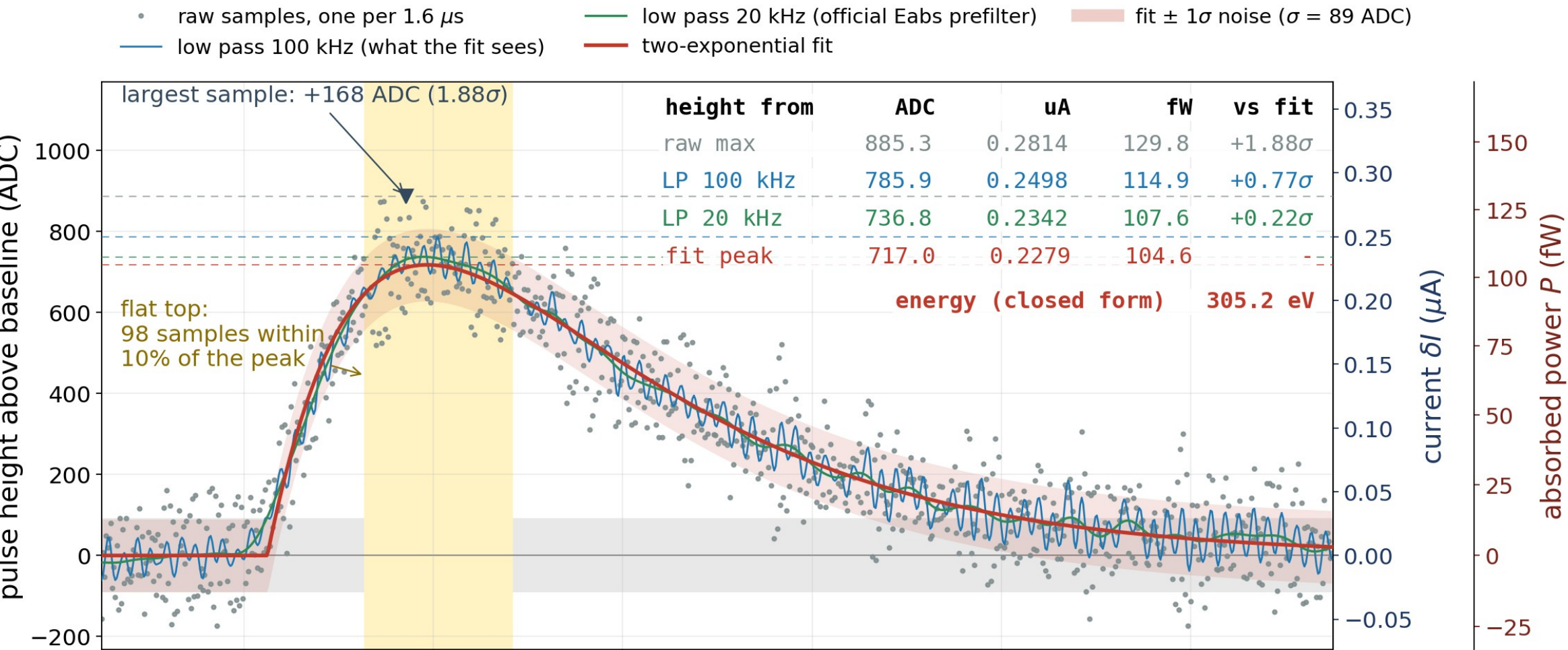
linear coefficient: $I_0(R_L - R_0) = (-18.069 \mu$ A) $\times$ (19.240 m Ω - 44.159 m Ω) = 0.450 μ V

quadratic coefficient: $c_2 R_L = 2 \times 19.240$ m Ω = 38.48 m Ω = 0.0385 Ω

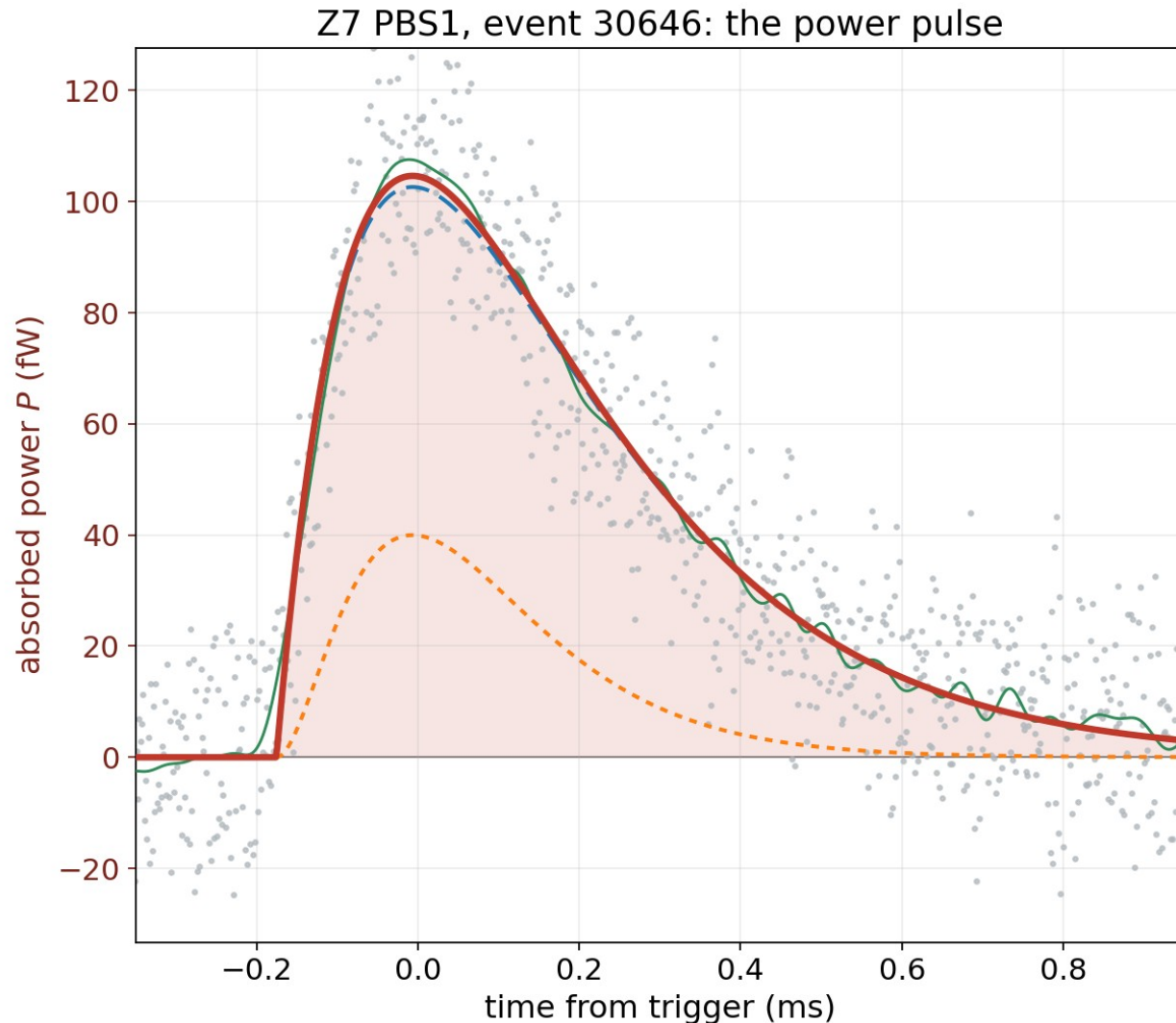
The two-exponential is fitted to the 100 kHz low-passed trace (blue), not to the raw samples; the raw samples are only drawn.

The height: take it from the fit, not from the largest sample

Z7 PBS1, event 30646: one pulse as ADC, as current and as power



The power pulse, and its area is the energy

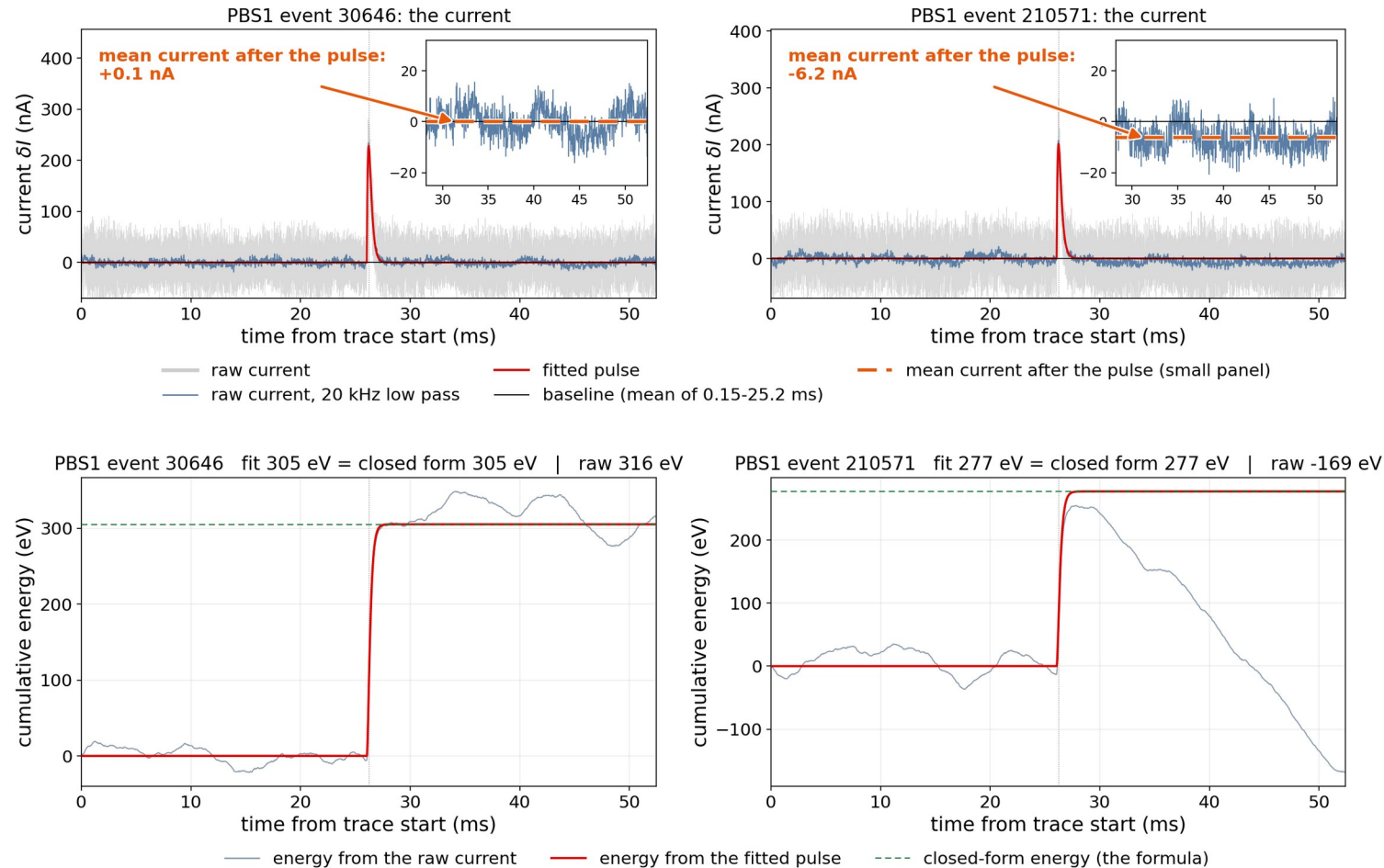


what is in the figure

- power from the fit, both terms
- - - the linear term alone
- - - the quadratic term alone, $\times 20$
- • • • • from the raw samples, for comparison
- from the 20 kHz trace, for comparison
- area = the energy absorbed here

**the quadratic term is 1.3 % of the energy,
so the power pulse has the shape of the cu**

Why we integrate the fit, not the raw data

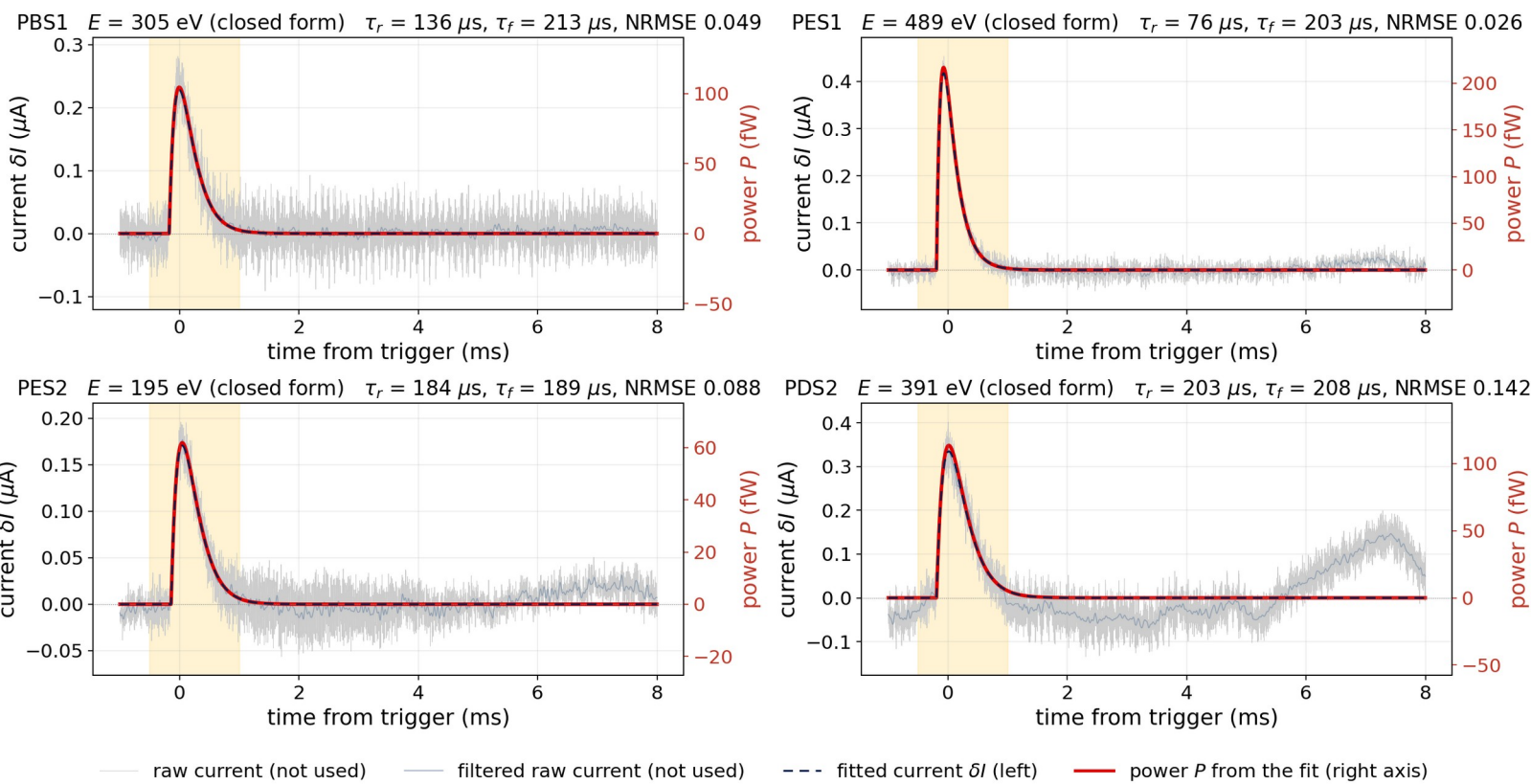


Top: the raw current of the two events; the small panel zooms in on the baseline after the pulse. Bottom: the energy accumulated from the raw current (grey) and from the fit (red).

Right: after the pulse the current sits 6 nA below its baseline — hidden in 28 nA of noise, yet over 24 ms it is -420 eV: raw -169 eV, fit 277 eV.

The same event in its other channels: they add up to 33 %

Z7 event 30646; four of the eleven channels



energy per channel, event 30646

side 1		2062 eV (60 %)
PAS1		303 eV
PBS1		305 eV
PCS1		227 eV
PDS1		378 eV
PES1		489 eV
PFS1		359 eV
side 2		1357 eV (40 %)
PAS2		267 eV
PBS2		261 eV
PCS2		243 eV
PDS2		391 eV
PES2		195 eV

total 3419 eV = 33.0 % of 10.37 keV

PFS2 is not read out on Z7, so side 2 has 5 channels
the channels also differ from each other; see the backup

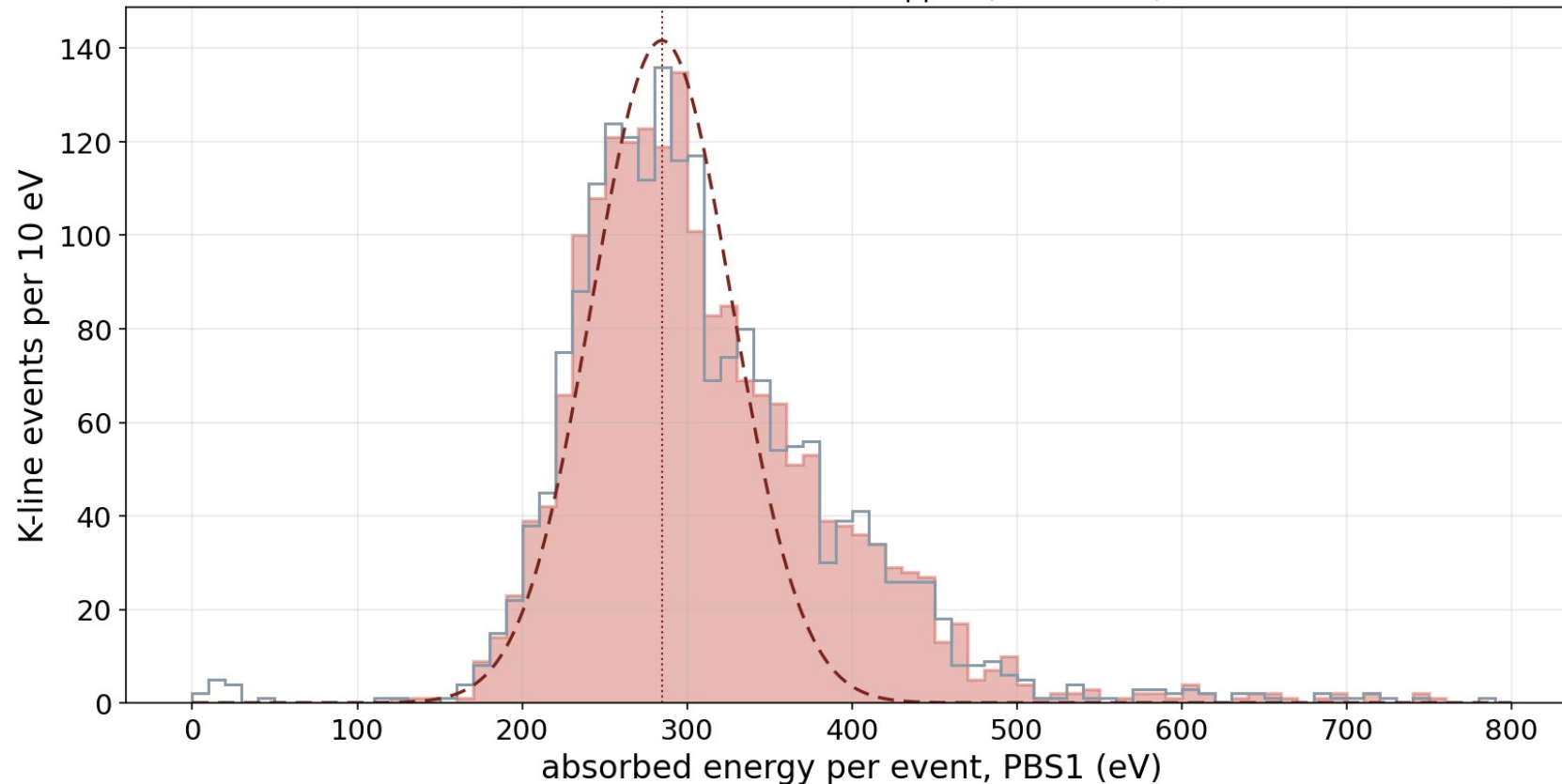
The channels do not share the energy equally: 60 % of it lands on side 1, and PES1 alone has 489 eV against 195 eV in PES2.

Summed over the eleven channels: 3419 eV = 33.0 % of 10.37 keV. PDS2 (bottom right) carries a slow swing.

Now every event: one channel gives a wide, lopsided distribution

PBS1, 1917 K-line events, one fit and one integral each

Z7 PBS1: energy of every K-line event (1917 events, 30 series)
two-exponential fit per event, closed-form power integral (method1)
fits with NRMSE > 0.2 dropped (290 events)



- official-window raw energy: $\mu = 284.2$ eV, $\sigma = 43.1$ eV (15.2%)
- fitted-pulse closed-form energy: $\mu = 284.8$ eV, $\sigma = 42.4$ eV (14.9%)
- Gaussian fit to the core

Red: from the fit. Grey: the official window on the raw trace. Same peak to 0.2 %.

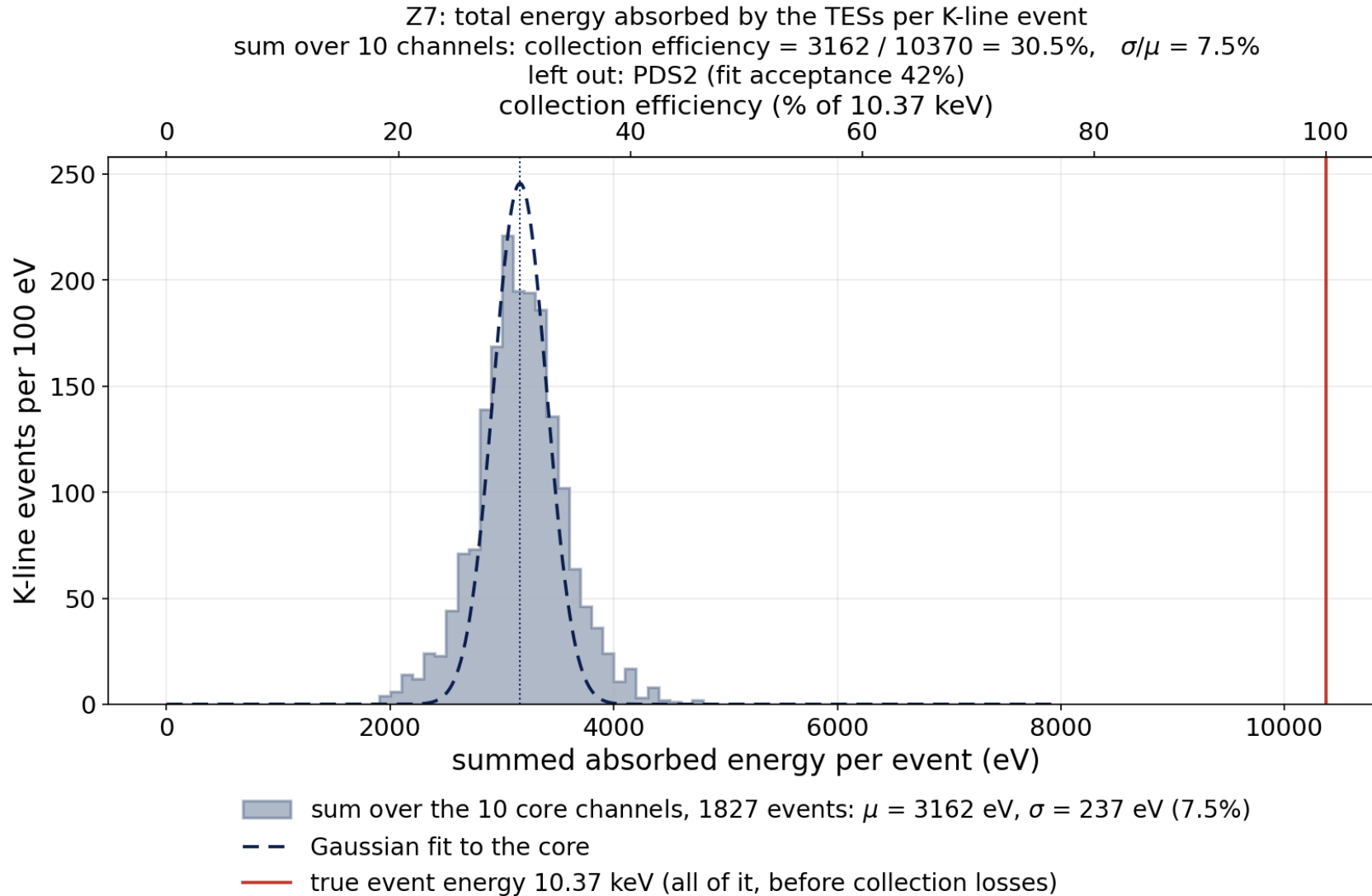
Peak 285 eV, width 15 % — for a line whose energy is fixed.

The tail to the right is the events that happened near this channel.

One channel's width is event position, not noise.

Summed over the channels: 30.5 %

the total energy the TESs absorbed per K-line event



Ten channels summed, 1827 events: 3162 eV = 30.5 %, and the width drops to 7.5 % — the position effect cancels.

PDS2 is left out: only 42 % of its events can be fitted, and those are the events that happened near it.

Collection efficiency:

30.5 % $\pm$ 2.3 % ($\sigma/\mu = 7.5$ %).

Red line: 10.37 keV — two thirds of the energy never reaches the TESs.

Summary, and what is still open

The current pulse becomes a power pulse through the TES equations; the area under the power is the energy.

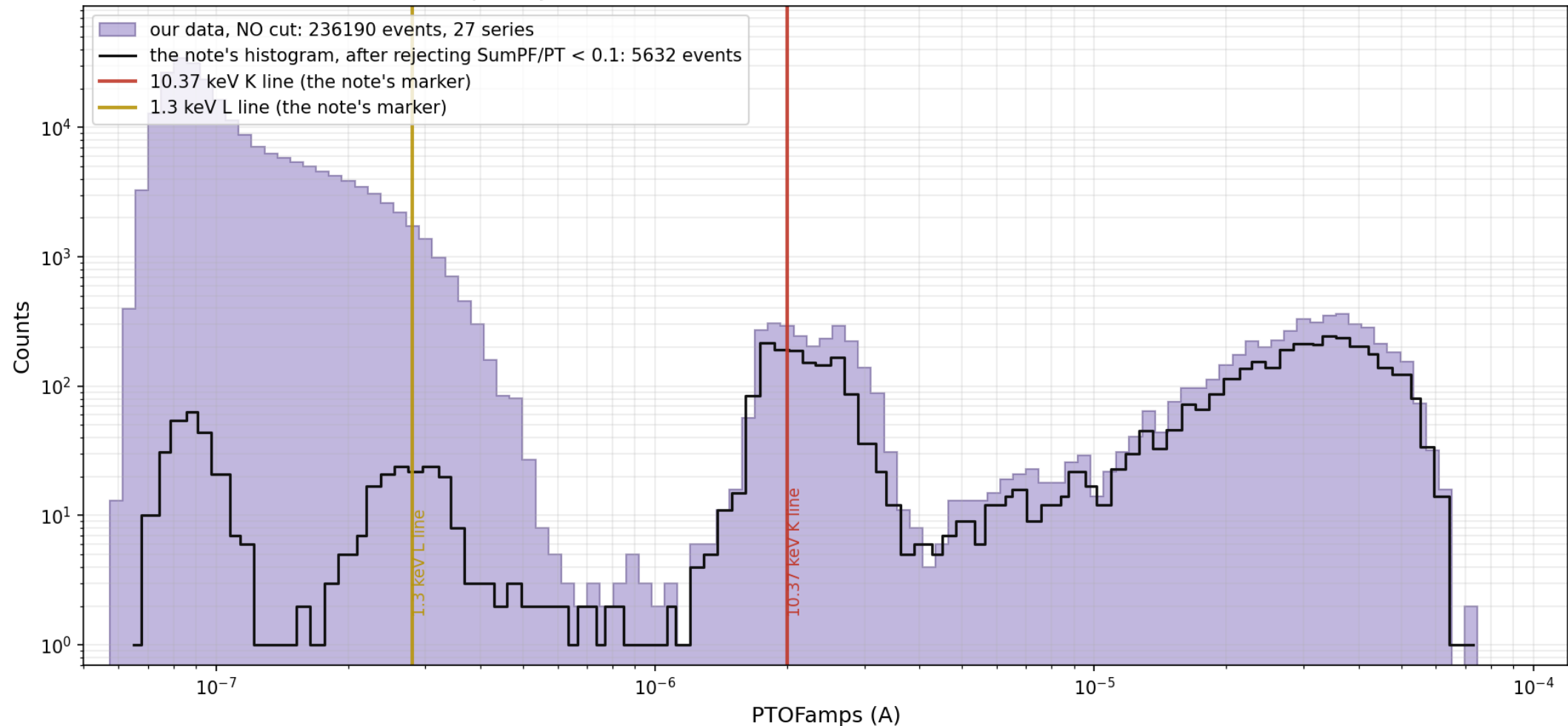
The energy is taken from the fitted pulse, so the noise at the peak and the baseline drift do not enter.

Z7 absorbs $30.5 \% \pm 2.3 \%$ ($\sigma/\mu = 7.5 \%$) of a 10.37 keV event in its TESs, summed over 10 channels. The CDMS note reports 26–41 % on the R37 CUTE tower.

Open: Z7 only · the bias of Z7 is not established (0 V assumed) · no dI/dV · PDS2 not included · 12.5 % of events fit in no channel.

Backup — the events

Z7: summed PTOFamps over 27 series, rebuilt from the Prompt processing with no cut applied
the note's version of the same plot rejects SumPF/PT below 0.1; the difference between the two curves is what that cut removes



Backup — the two Methods of the CDMS note

$\delta P = I_0(R_L - R_0) \cdot \delta I + c_2 \cdot R_L \cdot (\delta I)^2$ with $c_2 = 2$ (Method 1, used here) and $c_2 = -1$ (Method 2)

The two differ only in the quadratic term: 1.7 % in energy (PBS1, median of 15 events) — far below the 15 % single-channel width and the 7.5 % width of the sum.

The inductor term of the full equations needs L and the loop gain from dI/dV , which do not exist for these series on MSI; it is neglected.

The official Eabs (trigger bin 16383, baseline bins 93..15758, 5-pole 20 kHz prefilter, window $-0.5/+1$ ms, same formula) is reproduced to 0.2 %; the grey outlines on the histogram slides are it.

Backup — how the formulas are derived

1. From the current to the absorbed power

heat balance of the TES

$$C \dot{T} = P(t) + P_J - P_{bath}$$

over the whole pulse $\int C \dot{T} dt = 0$; the change in bath heat flow is not included

$$E = \int P dt = - \int \delta P_J dt$$

the circuit; at the bias point $V = I_0(R_L + R_0)$

$$V_{TES} = V - IR_L - L \dot{I}, \quad P_J = IV_{TES}$$

put $I = I_0 + \delta I$ and subtract the value at the bias point

$$\delta P_J = V \delta I - R_L(2I_0 \delta I + (\delta I)^2) - L I \delta \dot{I}$$

integrate: the L term gives $L[I_0 \delta I + (\delta I)^2/2]$, which is 0 at both ends

$$E = \int \delta P dt, \quad \delta P = I_0(R_L - R_0) \delta I + c_2 R_L (\delta I)^2$$

$c_2 = 1$ from this derivation, 2 in Method 1 (used here), -1 in Method 2

2. Energy of the fitted pulse

the fit, with t counted from the pulse start t_0

$$\delta I(t) = A [e^{-t/\tau_f} - e^{-t/\tau_r}]$$

the only integral needed

$$\int_0^\infty e^{-t/\tau} dt = \tau$$

linear term

$$\int \delta I dt = A(\tau_f - \tau_r)$$

quadratic term: expand the square, integrate each exponential

$$\int (\delta I)^2 dt = A^2 \left[\frac{\tau_f + \tau_r}{2} - \frac{2\tau_f \tau_r}{\tau_f + \tau_r} \right]$$

so, with $c_2 = 2$

$$E = I_0(R_L - R_0) A(\tau_f - \tau_r) + 2R_L A^2 \left[\frac{\tau_f + \tau_r}{2} - \frac{2\tau_f \tau_r}{\tau_f + \tau_r} \right]$$

PBS1, event 30646: $A = 1.39 \mu A$, $\tau_r = 136 \mu s$, $\tau_f = 213 \mu s \Rightarrow E = 305.2 \text{ eV}$

3. Symbols (Z7 PBS1 values) and assumptions

I_b : bias current R_{sh} : shunt resistor, 5.00 m Ω $V = I_b R_{sh}$: bias voltage, $-1.146 \mu V$ R_p : parasitic resistance, 14.24 m Ω

$R_L = R_p + R_{sh}$: load resistance, 19.24 m Ω R_0 : TES resistance at the bias point, 44.16 m Ω $I_0 = V/(R_L + R_0)$: TES current, $-18.07 \mu A$

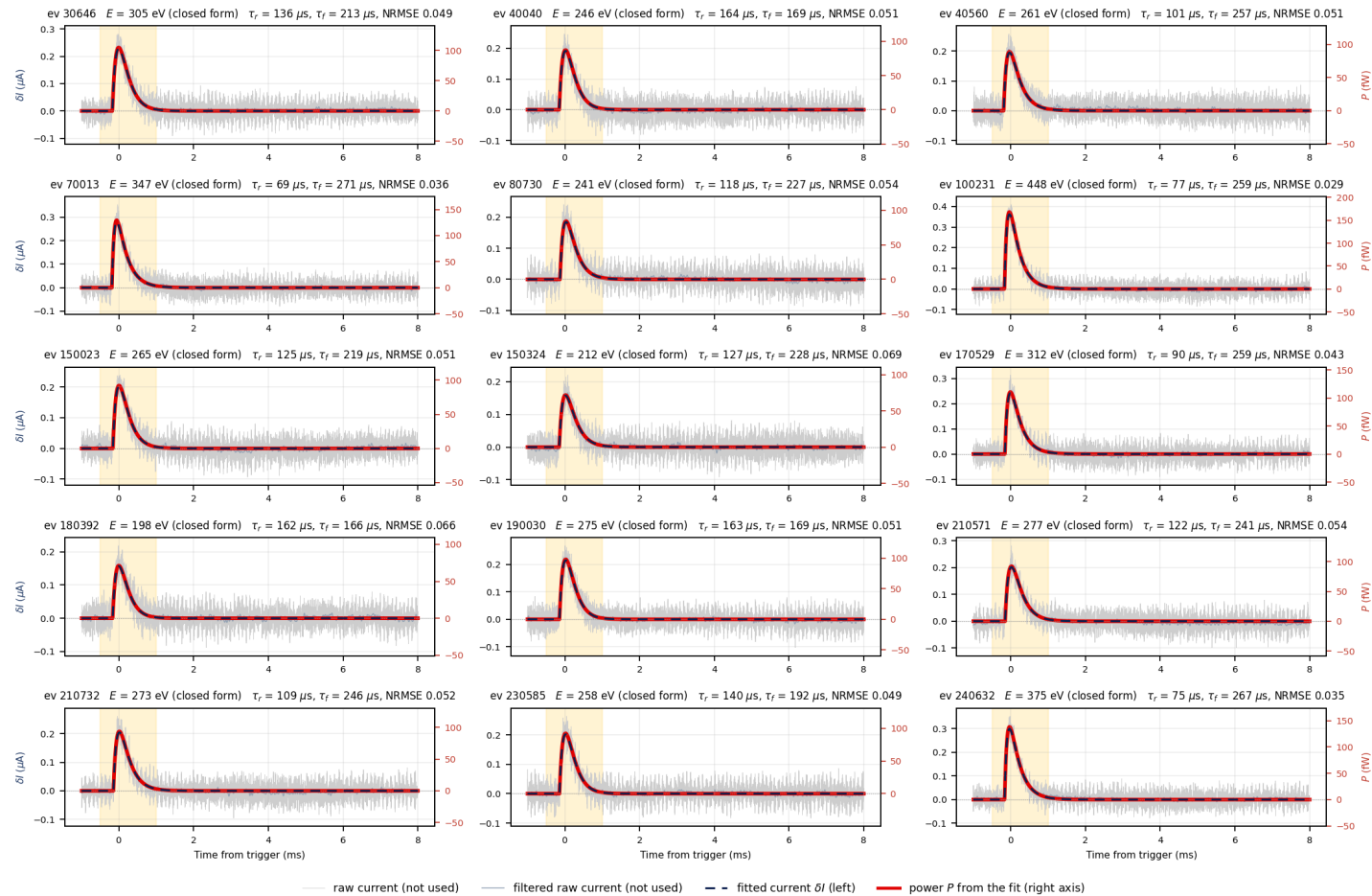
L : inductance (unknown) δI : current change A : amplitude τ_r, τ_f : rise and fall time constants t_0 : pulse start

Assumptions: detector at 0 V (no NTL effect); the heat flow to the bath is not included;

the L term cancels only after integration, so $\delta P(t)$ is not the exact power at each moment; only its integral is used

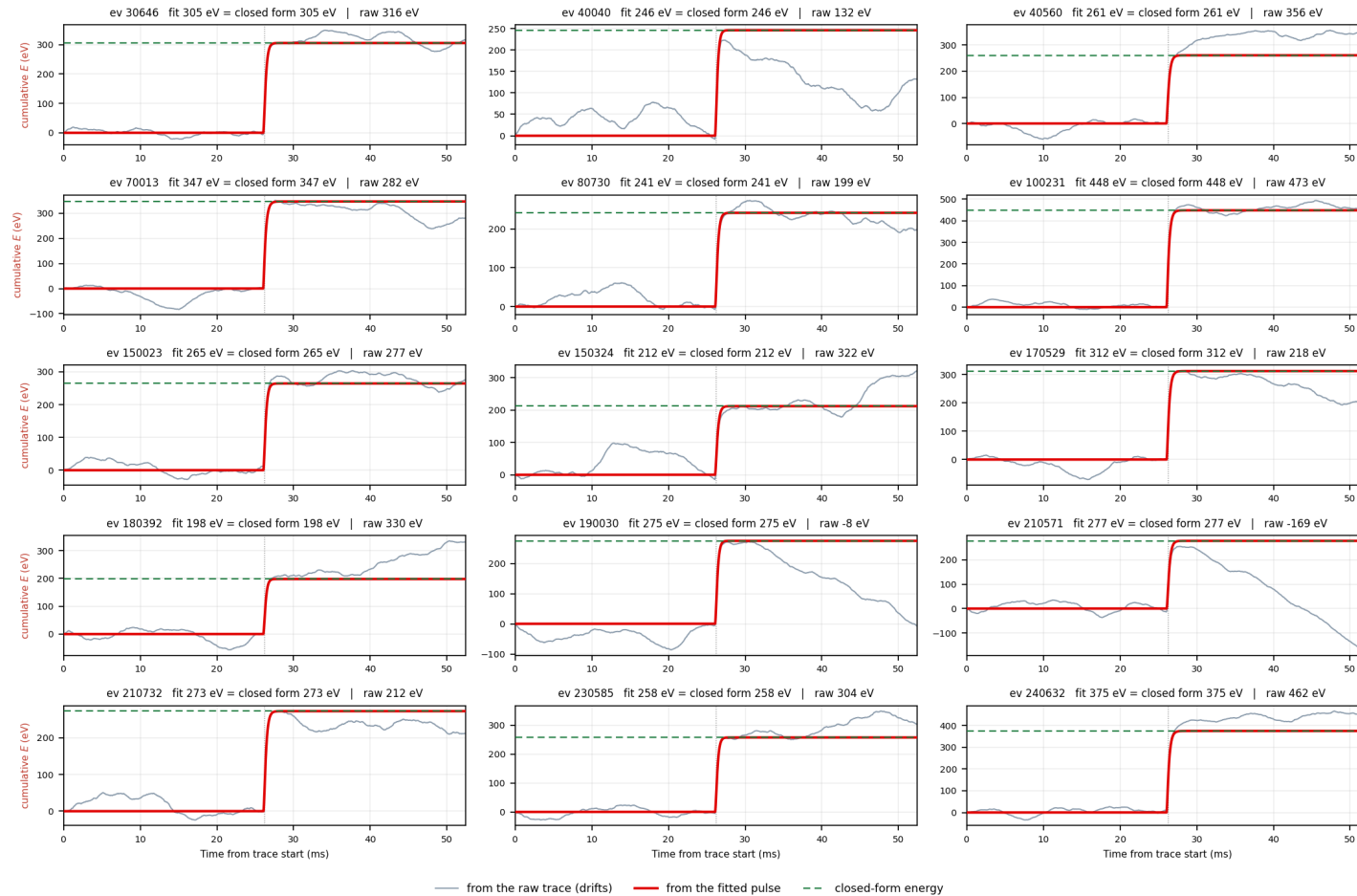
Backup — 15 events, fitted current and power (PBS1)

Z7 PBS1: fitted current pulse (dashed navy) and the power pulse it implies (red)
 Z7 24260617_063934, K-line events | method1: $P(t) = 10(R/L-R_0) \cdot dI(t) + 2 \cdot R \cdot L \cdot (dI(t))^2$ | bias point per channel from detectorConfig, ADC/A = 3.145728e+09
 the current is the two-exponential FIT, refitted directly in amperes (model $A[\exp(-t/\tau_r) - \exp(-t/\tau_f)]$), 5 free params, pretrigger free in 16383+-3000; the raw trace is drawn only for reference
 integration runs over the whole trace, 0 to 52.43 ms; for the fitted pulse this equals the closed-form integral, so the energy no longer depends on the window



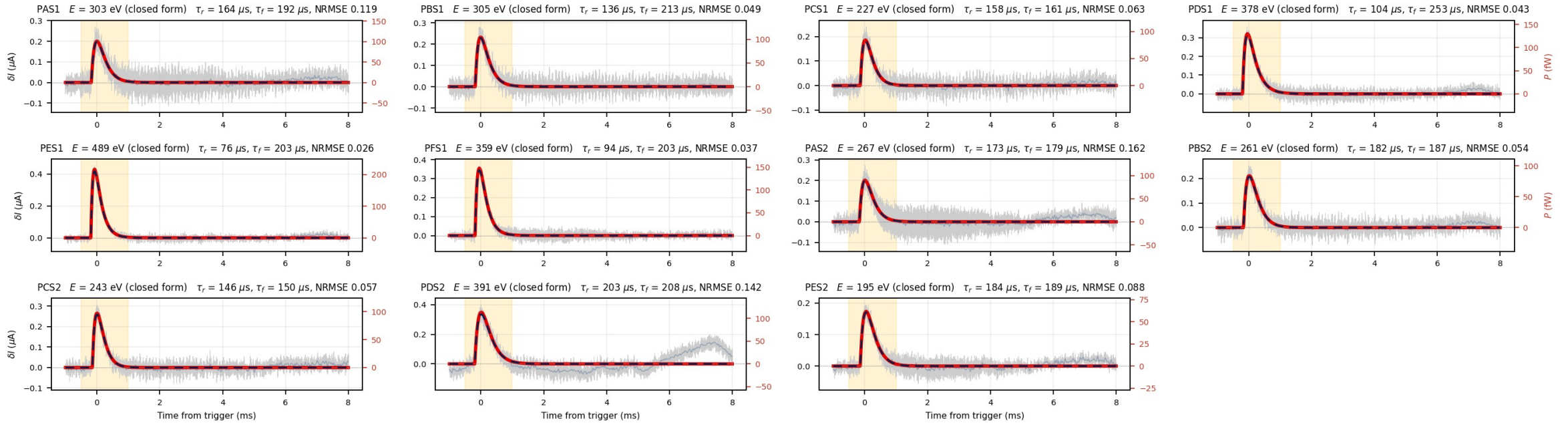
Backup — 15 events, cumulative energy (PBS1)

Z7 PBS1: integrated power = absorbed energy, fitted pulse vs raw trace
 Z7 24260617_063934, K-line events | method1: $P(t) = 10(R_L R_0) \cdot d(t) + 2 \cdot R_L \cdot L \cdot (d(t))^2$ | bias point per channel from detectorConfig, ADC/A = 3.145728e+09
 the current is the two-exponential FIT, refitted directly in amperes (model $A[\exp(-t/\tau_f) - \exp(-t/\tau_r)]$), 5 free params, pretrigger free in 16383+-3000; the raw trace is drawn only for reference
 integration runs over the whole trace, 0 to 52.43 ms; for the fitted pulse this equals the closed-form integral, so the energy no longer depends on the window



Backup — event 30646, all channels

Z7 event 30646, all channels: fitted current and power
 sum over channels from the fits: 3419 eV = 33.0% of 10.37 keV (official-window raw sum 3397 eV = 32.8%)
 Z7 24260617_063934, K-line events | method1: $P(t) = I_0(R_{L-R0}) * dl(t) + 2 * R_{L-R0} * L * (dl(t))^2$ | bias point per channel from detectorConfig, ADC/A = 3.145728e+09
 the current is the two-exponential FIT, refitted directly in amperes (model $A[\exp(-t/\tau_f) - \exp(-t/\tau_r)]$, 5 free params, pretrigger free in 16383+-3000); the raw trace is drawn only for reference
 integration runs over the whole trace, 0 to 52.43 ms; for the fitted pulse this equals the closed-form integral, so the energy no longer depends on the window



Backup — energy distribution of every channel

Z7: energy of every K-line event, one panel per channel (30 series)

each entry is one event: its trace fitted with the two-exponential and the power integrated in closed form, method1 quadratic

the dashed curve is a Gaussian fit to the core, so its centre is the PEAK; these distributions are right skewed, so peaks must not be added (the plain means do add, to 3710 eV = 35.8% of the 10.37 keV K line)

